

Development Roadmap

Phased implementation plan, deployment architecture, and portfolio completion targets

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1. Purpose

This roadmap defines the recommended implementation order for FleetFlow. It is intentionally incremental so the project can remain stable while new microservices and workflows are added.

Legend:

-  Implemented / established
-  In progress / partially implemented
-  Planned

2. Phase 1 — Foundation

-  Create FleetFlow monorepo structure
-  Create Next.js web application
-  Establish TypeScript frontend
-  Establish authentication service
-  Connect MongoDB to backend services
-  Establish JWT authentication
-  Establish user roles
-  Create shared frontend authentication context
-  Create login/register flows
-  Establish light/dark theme
-  Create dashboard layout

3. Phase 2 — Vehicle Management

-  Vehicle domain/service foundation
-  Vehicle creation
-  Vehicle listing
-  Vehicle retrieval
-  Vehicle update
-  Vehicle deactivation
-  Vehicle status management
-  Vehicle filtering/search foundation
-  Vehicle dashboard UI
-  Vehicle service/history UI

4. Phase 3 — Vehicle Requests

-  Vehicle request page foundation (*in progress*)
-  Request creation API finalized
-  Request lifecycle/state machine finalized
-  Manager request review
-  Request approval/rejection
-  Vehicle assignment
-  Customer request history
-  Driver assignment workflow
-  Request notifications

5. Phase 4 — Fleet Operations

-  Driver management service
-  Driver profiles
-  Driver status
-  Vehicle assignment
-  Fleet availability management
-  Operational dashboard
-  Driver-specific workflows

6. Phase 5 — Maintenance

-  Vehicle service/history foundation (*in progress*)
-  Dedicated maintenance domain
-  Maintenance schedules
-  Maintenance status
-  Maintenance costs
-  Upcoming maintenance
-  Maintenance reminders
-  Maintenance reporting

7. Phase 6 — Notifications

-  Notification service
-  In-app notification model
-  Read/unread state
-  Request notifications

- ☐ Assignment notifications
- ☐ Maintenance notifications
- ☐ Notification preferences

8. Phase 7 — Analytics & Reporting

- ☐ Fleet utilization metrics
- ☐ Vehicle status analytics
- ☐ Request analytics
- ☐ Maintenance analytics
- ☐ Driver metrics
- ☐ Date filtering
- ☐ Exportable reports

9. Phase 8 — Production Hardening

- ☐ Automated unit tests
- ☐ API integration tests
- ☐ Frontend tests
- ☐ End-to-end tests
- ☐ Centralized error handling
- ☐ Structured logging
- ☐ Rate limiting
- ☐ Security review
- ☐ Dependency audit
- ☐ Database indexes
- ☐ Backup strategy
- ☐ Monitoring
- ☐ Health-check endpoints

10. Phase 9 — Deployment

Target deployment architecture:

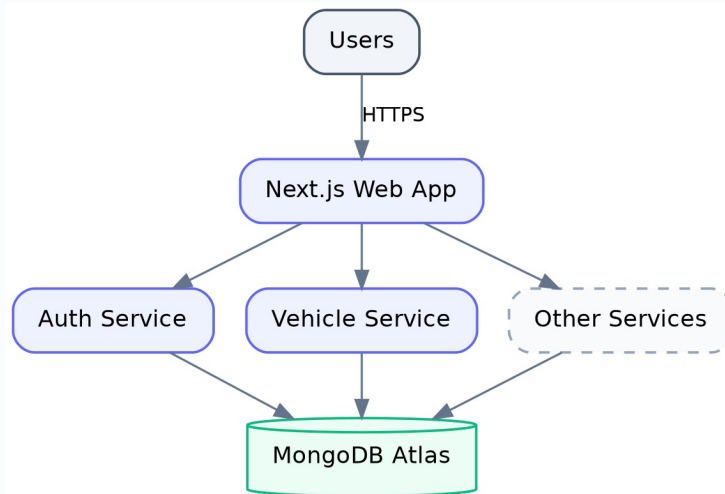


Figure 1 — Target deployment architecture

Deployment providers are implementation choices and should be finalized based on cost, reliability, and project requirements.

11. Definition of Done

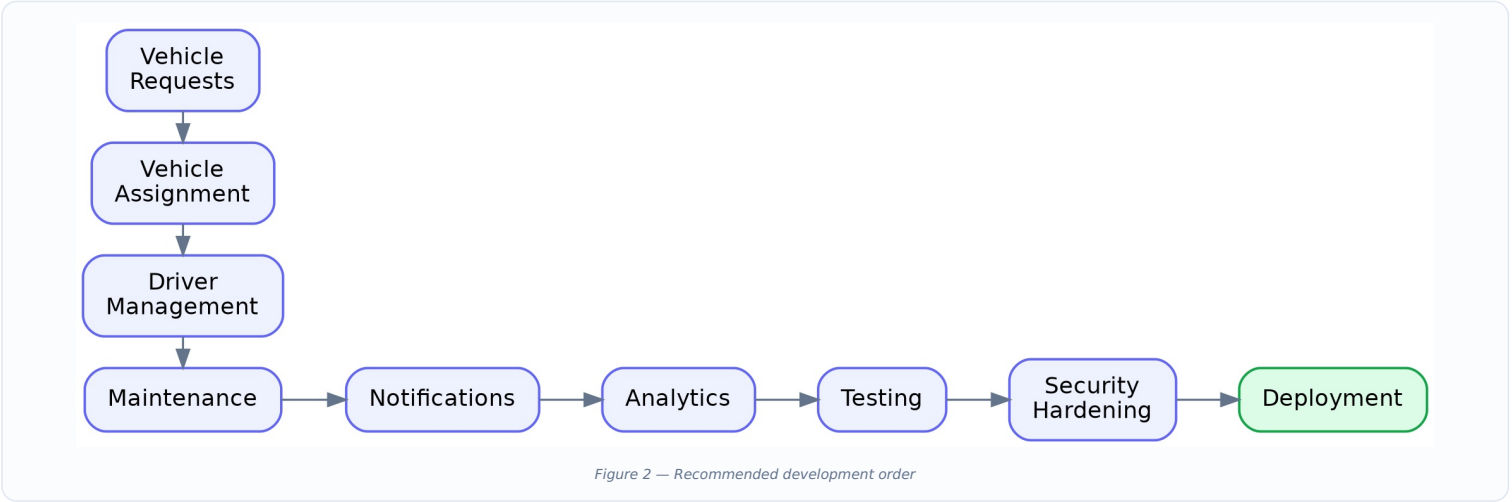
A feature should not be marked complete merely because the UI exists.

A feature is done when:

- ☐ UI is implemented
- ☐ Backend endpoint exists where required
- ☐ Validation exists
- ☐ Authorization exists
- ☐ Database persistence works
- ☐ Loading states work
- ☐ Error states work
- ☐ Empty states work
- ☐ Responsive behavior is verified
- ☐ Relevant API documentation is updated
- ☐ Tests are added when test infrastructure supports them
- ☐ Git commit clearly describes the change

12. Recommended Development Order

The recommended next sequence is:



This order keeps the core operational workflow coherent before adding secondary platform features.

13. Portfolio Completion Target

Before presenting FleetFlow as a finished portfolio project, the project should demonstrate:

- Microservice architecture
- Next.js + TypeScript frontend
- Authentication
- JWT
- RBAC
- MongoDB
- REST APIs
- Input validation
- Vehicle management
- Vehicle request workflow
- Vehicle service/history
- Driver/assignment workflow
- Maintenance workflow
- Notifications
- Analytics
- Automated testing
- Deployment
- Professional documentation